Generalized oddtowns and eventowns

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Q: What is the maximum m?

Notations

- $[n] = \{1, 2, \ldots, n\}$.
- $2^{[n]}$ denotes the set of all subsets of $[n]$.
- $\mathcal{F} = \{F_1, \ldots, F_m\}$ is a set family.

Oddtown and eventown

- $\mathcal{F}$ is an oddtown: $\begin{cases} |A| \equiv 1 \pmod 2, \\ |A \cap B| \equiv 0 \pmod 2. \end{cases}$

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- $\mathcal{F}$ is an eventown: $\begin{cases} |A| \equiv 0 \pmod 2, \\ |A \cap B| \equiv 0 \pmod 2. \end{cases}$

  e.g. $\mathcal{F} = \big\{\{1,2\},\{3,4\},\{5,6,7,8\}\big\}$.

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$$\left| \bigcap_{j=1}^{i} F_j \right| \equiv \alpha_i \pmod 2 \quad \forall\, 1 \leq i \leq k,\ \forall \text{ distinct } F_1, \ldots, F_i \in \mathcal{F}.$$

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- $f_{\boldsymbol{\alpha}}(n) := \max\{|\mathcal{F}| : \mathcal{F} \subseteq 2^{[n]} \text{ and } \mathcal{F} \text{ is an } \boldsymbol{\alpha}\text{-town}\}$.

  e.g. $f_{(1,0)}(n) = n$, $f_{(0,0)}(n) = 2^{\lfloor n/2 \rfloor}$.

- Johnston–O'Neill (2025):
  - figured out $f_{\boldsymbol{\alpha}}(n)$ for every $\boldsymbol{\alpha} \in \mathbb{F}_2^3 \cup \mathbb{F}_2^4$
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- Wei–Zhang–Ge (2025):
  - established $f_{\boldsymbol{\alpha}}(n) \sim (t!n)^{\frac{1}{t}}$ for $\boldsymbol{\alpha} = (\underbrace{*, \ldots, *, 1}_{\text{short}}, \underbrace{0, \ldots, 0}_{\text{long}})$
  - $\#$ of $\boldsymbol{\alpha} \in \mathbb{F}_2^k$ s.t. $f_{\boldsymbol{\alpha}}(n) = \Omega\left(\sqrt{n}\right)$ is at most 128

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Problem 2 (Wei–Zhang–Ge, 2025)

For a fixed k*, determine the asymptotics of* $f_{\boldsymbol{\alpha}}(n)$ *for all* $\boldsymbol{\alpha} \in \mathbb{F}_2^k$.

Our result

Theorem 2.1 (Dong–Ouyang–W., 2026+)

$$f_{\boldsymbol{\alpha}}(n) = \begin{cases} 2^{\lfloor n/2 \rfloor} & \text{if } \mathrm{lv}(\boldsymbol{\alpha}) = 0 \text{ and } \boldsymbol{\alpha} = (0, \ldots, 0), \\ 2^{\lfloor (n-1)/2 \rfloor} & \text{if } \mathrm{lv}(\boldsymbol{\alpha}) = 0 \text{ and } \boldsymbol{\alpha} = (1, \ldots, 1), \\ \big(1 + o(1)\big) \cdot (t!)^{1/t} \cdot n^{1/t} & \text{if } \mathrm{lv}(\boldsymbol{\alpha}) = t > 0 \text{ and } \mathrm{grd}(\boldsymbol{\alpha}) = 1, \\ \big(1 + o(1)\big) \cdot (t!/2)^{1/t} \cdot n^{1/t} & \text{if } \mathrm{lv}(\boldsymbol{\alpha}) = t > 0 \text{ and } \mathrm{grd}(\boldsymbol{\alpha}) = 2. \end{cases}$$

$$\gamma = \begin{pmatrix} \gamma^1 \\ \gamma^2 \\ \gamma^3 \\ \gamma^4 \\ \gamma^5 \\ \gamma^6 \\ \vdots \end{pmatrix} = \begin{pmatrix} \binom{-1}{0} & \binom{-1}{1} & \binom{-1}{2} & \binom{-1}{3} & \binom{-1}{4} & \binom{-1}{5} & \cdots \\ \binom{0}{0} & \binom{0}{1} & \binom{0}{2} & \binom{0}{3} & \binom{0}{4} & \binom{0}{5} & \cdots \\ \binom{-2}{0} & \binom{-2}{1} & \binom{-2}{2} & \binom{-2}{3} & \binom{-2}{4} & \binom{-2}{5} & \cdots \\ \binom{1}{0} & \binom{1}{1} & \binom{1}{2} & \binom{1}{3} & \binom{1}{4} & \binom{1}{5} & \cdots \\ \binom{-3}{0} & \binom{-3}{1} & \binom{-3}{2} & \binom{-3}{3} & \binom{-3}{4} & \binom{-3}{5} & \cdots \\ \binom{2}{0} & \binom{2}{1} & \binom{2}{2} & \binom{2}{3} & \binom{2}{4} & \binom{2}{5} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

- $(\gamma^{2i-1})_j = \binom{-i}{j-1} \pmod 2, \qquad (\gamma^{2i})_j = \binom{i-1}{j-1} \pmod 2.$

Every $\boldsymbol{\alpha} \in \mathbb{F}_2^k$ has a unique *canonical decomposition*

$$\boldsymbol{\alpha} = c_1 \cdot (\boldsymbol{\gamma}^1)_{[k]} + \cdots + c_k \cdot (\boldsymbol{\gamma}^k)_{[k]}.$$

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$$\begin{cases} \mathrm{lv}(\boldsymbol{\alpha}) = 2 \\ \mathrm{grd}(\boldsymbol{\alpha}) = 1 \end{cases} \implies f_{\boldsymbol{\alpha}}(n) = (1 + o(1)) \cdot (t!)^{1/t} \cdot n^{1/t} = (1 + o(1)) \cdot \sqrt{2n}$$

Main lemma

Lemma 3 (Johnston–O'Neill, Partition–Sum Lemma)

Let $x, y \in \mathbb{N}_+$. *If* $\min\{f_{\boldsymbol{\alpha}}(x), f_{\boldsymbol{\beta}}(y)\} > k$, *then*

$$f_{\boldsymbol{\alpha}+\boldsymbol{\beta}}(x+y) \geq \min\{f_{\boldsymbol{\alpha}}(x), f_{\boldsymbol{\beta}}(y)\}.$$

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Corollary 2.2

Let $x_1, \ldots, x_\ell \in \mathbb{N}_+$. *If* $\min\{f_{\boldsymbol{\alpha}_1}(x_1), \ldots, f_{\boldsymbol{\alpha}_\ell}(x_\ell)\} > k$, *then*

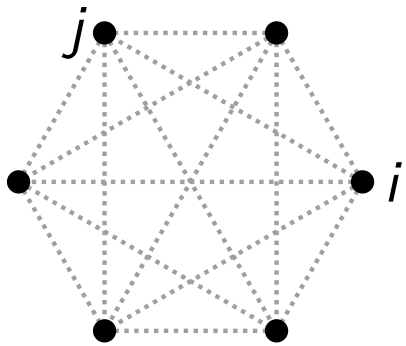
$$f_{\boldsymbol{\alpha}_1+\cdots+\boldsymbol{\alpha}_\ell}(x_1+\cdots+x_\ell) \geq \min\{f_{\boldsymbol{\alpha}_1}(x_1), \ldots, f_{\boldsymbol{\alpha}_\ell}(x_\ell)\}.$$

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- (Hyper)star constructions, which is good for $f_{\gamma^i_{[k]}}(n)$.

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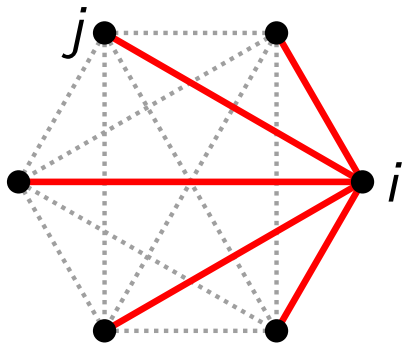
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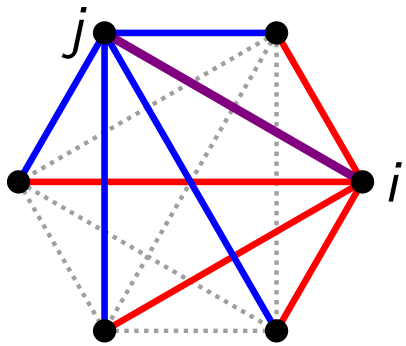
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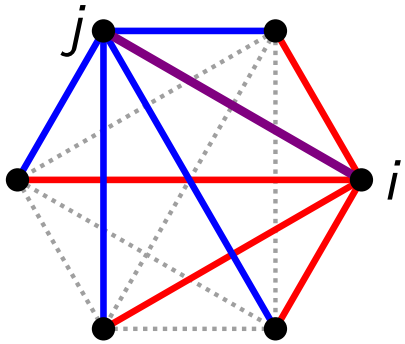
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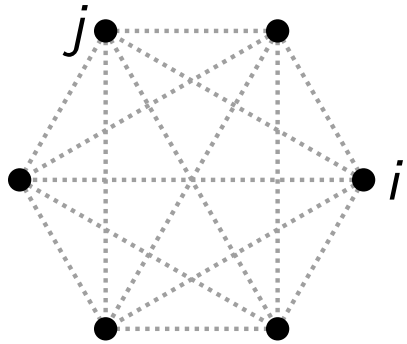
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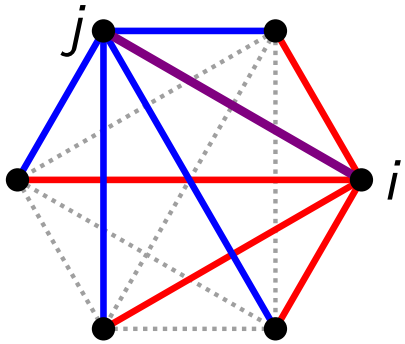
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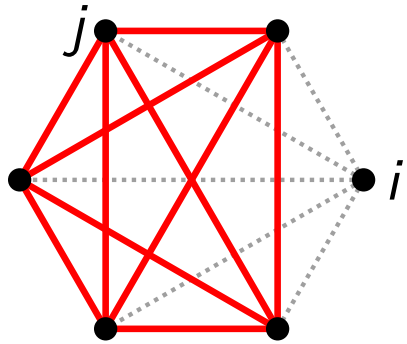
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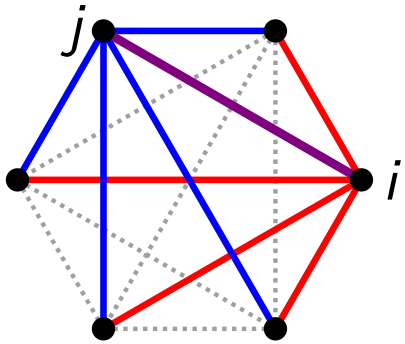
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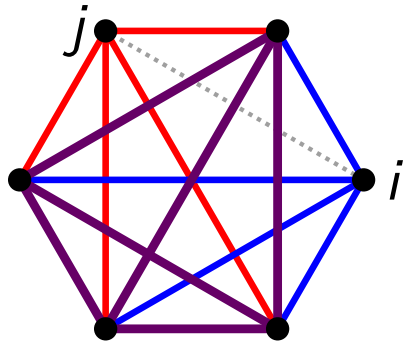
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Lower bound on $f_{\alpha}(n)$

- (Hyper)star constructions, which is good for $f_{\gamma^i_{[k]}}(n)$.
- Apply the Partition–Sum Lemma to $\boldsymbol{\alpha} = \gamma^{i_1}_{[k]} + \cdots + \gamma^{i_\ell}_{[k]}$:

$$f_{\alpha}(n) \geq \min\left\{ f_{\gamma^{i_1}_{[k]}}(n_1), \ldots, f_{\gamma^{i_\ell}_{[k]}}(n_\ell) \right\}, \qquad n_1 + \cdots + n_\ell = n.$$

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Upper bound on $f_{\boldsymbol{\alpha}}(n)$

- Locate a subclass of $\boldsymbol{\gamma}$ for which $\overline{f_{\boldsymbol{\gamma}}}(n)$ is well-understood
- (KEY) Choose $\boldsymbol{\beta}$ s.t. $\boldsymbol{\gamma} = \boldsymbol{\alpha} + \boldsymbol{\beta}$ and $\underline{f_{\boldsymbol{\beta}}}(n) \gg \overline{f_{\boldsymbol{\gamma}}}(n)$
- Apply the Partition–Sum Lemma to $\boldsymbol{\gamma} = \boldsymbol{\alpha} + \boldsymbol{\beta}$:

$$f_{\boldsymbol{\gamma}}(n+m) \geq \min\{f_{\boldsymbol{\alpha}}(n), f_{\boldsymbol{\beta}}(m)\} \quad (m = o(n))$$

- Make $\underline{f_{\boldsymbol{\beta}}}(m) \gg \overline{f_{\boldsymbol{\gamma}}}(n+m)$ so that we obtain $\overline{f_{\boldsymbol{\alpha}}}(n)$

Future direction

- **$\boldsymbol{\alpha} \pmod p$-town:** For $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_k) \in \mathbb{F}_p^k$, call $\mathcal{F}$ an $\boldsymbol{\alpha}$-*town* if

$$\left| \bigcap_{j=1}^{i} F_j \right| \equiv \alpha_i \pmod p \quad \forall\, 1 \le i \le k,\ \forall\, \text{distinct } F_1, \ldots, F_i \in \mathcal{F}.$$

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Problem 4

Understand $f_{\boldsymbol{\alpha}}(n)$ for $\boldsymbol{\alpha} \in \mathbb{F}_p^k$.

Thanks for your attention!